

Lemma: Let $\{\vec{x}_1, \dots, \vec{x}_m\}$ be a set of non-zero orthogonal vectors. Then $\{\vec{x}_1, \dots, \vec{x}_m\}$ is linearly independent.

Proof. Suppose $a_1 \vec{x}_1 + \dots + a_m \vec{x}_m = \vec{0}$. Then for each $j=1, \dots, m$
 $0 = (\vec{0}, \vec{x}_j) = \left(\sum_{i=1}^m a_i \vec{x}_i, \vec{x}_j \right) = \sum_{i=1}^m a_i (\vec{x}_i, \vec{x}_j) = a_j \underbrace{(\vec{x}_j, \vec{x}_j)}_{\neq 0} \Rightarrow a_j = 0$ $\square$

Lemma: Let $\{\vec{x}_1, \dots, \vec{x}_m\}$ be an orthogonal set of non-zero vectors.

Let $\vec{y} \in \text{Span}\{\vec{x}_1, \dots, \vec{x}_m\}$. Then $\vec{y} = \sum_{i=1}^m \frac{(\vec{y}, \vec{x}_i)}{(\vec{x}_i, \vec{x}_i)} \vec{x}_i$.

Proof: Suppose $\vec{y} = \sum_{i=1}^n a_i \vec{x}_i$.

Then $(\vec{y}, \vec{x}_j) = \left(\sum_{i=1}^n a_i \vec{x}_i, \vec{x}_j \right)$ for each $j=1, \dots, m$
 $\downarrow$ Note linear w.r.t the first component.

$$\Leftrightarrow (\vec{y}, \vec{x}_j) = \sum_{i=1}^n a_i (\vec{x}_i, \vec{x}_j) \text{ for each } j=1, \dots, m$$

$$\Leftrightarrow (\vec{y}, \vec{x}_j) = a_j (\vec{x}_j, \vec{x}_j) \text{ for each } j=1, \dots, m$$

$$\Leftrightarrow a_j = \frac{(\vec{y}, \vec{x}_j)}{(\vec{x}_j, \vec{x}_j)} \text{ for each } j=1, \dots, m.$$

Corollary: Let $\{\vec{x}_1, \dots, \vec{x}_m\}$ be a set of orthonormal vectors. Let $\vec{y} \in \text{Span}\{\vec{x}_1, \dots, \vec{x}_m\}$

$$\text{Then } \vec{y} = \sum_{j=1}^m (\vec{y}, \vec{x}_j) \vec{x}_j.$$

Lemma: Let $\{\vec{x}_1, \dots, \vec{x}_k\} \subseteq V$ be a set of non-zero orthogonal vectors.

Then $\forall \vec{y} \in V$: $\vec{y} - \sum_{i=1}^k \frac{(\vec{y}, \vec{x}_i)}{(\vec{x}_i, \vec{x}_i)} \vec{x}_i$ is orthogonal to $\vec{x}_j$ for all $j=1, \dots, k$.

Proof: For each $j=1, \dots, k$: $\left(\vec{y} - \sum_{i=1}^k \frac{(\vec{y}, \vec{x}_i)}{(\vec{x}_i, \vec{x}_i)} \vec{x}_i, \vec{x}_j \right) = (\vec{y}, \vec{x}_j) - \sum_{i=1}^k \frac{(\vec{y}, \vec{x}_i)}{(\vec{x}_i, \vec{x}_i)} (\vec{x}_i, \vec{x}_j)$
 $\downarrow$ $\forall i \neq j, (\vec{x}_i, \vec{x}_j) = 0$
 $= (\vec{y}, \vec{x}_j) - \frac{(\vec{y}, \vec{x}_j)}{(\vec{x}_j, \vec{x}_j)} (\vec{x}_j, \vec{x}_j) = 0.$ $\square$

Theorem: Let $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ be a linearly independent set in an inner product space V . Then there exists a set of non-zero orthogonal vectors $\{\vec{y}_1, \dots, \vec{y}_m\}$ such that $\text{Span}\{\vec{x}_1, \dots, \vec{x}_k\} = \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\}$ for all $k=1, \dots, m$.

Proof. By induction on k . (Need to construct non-zero vectors $\{\vec{y}_1, \dots, \vec{y}_k\}$ such that:
 (1) $\{\vec{y}_1, \dots, \vec{y}_k\}$ is orthogonal
 (2) $\text{Span}\{\vec{x}_1, \dots, \vec{x}_k\} = \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\}$.)

Base Case: when $k=1$, take $\vec{y}_1 = \vec{x}_1$. Then $\{\vec{y}_1\}$ is non-zero orthogonal and $\text{Span}\{\vec{x}_1\} = \text{Span}\{\vec{y}_1\}$.

Inductive steps. Suppose we have constructed a set of non-zero vectors $\{\vec{y}_1, \dots, \vec{y}_{k-1}\}$ such that $\{\vec{y}_1, \dots, \vec{y}_{k-1}\}$ is orthogonal and $\text{Span}\{\vec{y}_1, \dots, \vec{y}_{k-1}\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_{k-1}\}$

$$\text{Take } \vec{y}_k = \vec{x}_k - \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i$$

$$\text{Then } \vec{y}_k \neq \vec{0} \quad (\text{b/c otherwise } \vec{x}_k = \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i \in \text{Span}\{\vec{y}_1, \dots, \vec{y}_{k-1}\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_{k-1}\})$$

$\Rightarrow \{\vec{x}_1, \dots, \vec{x}_k\}$ is linearly dependent, contradiction)

$$1) \quad \text{As } \text{Span}\{\vec{y}_1, \dots, \vec{y}_{k-1}\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_{k-1}\}$$

$$\vec{y}_k = \vec{x}_k - \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i \in \text{Span}\{\vec{x}_1, \dots, \vec{x}_{k-1}, \vec{x}_k\}$$

$$\in \text{Span}\{\vec{y}_1, \dots, \vec{y}_{k-1}\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_{k-1}\}$$

$$\Rightarrow \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\} \subseteq \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\}$$

$$\vec{x}_k = \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i + \vec{y}_k \in \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\}$$

$$\Rightarrow \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\} \subseteq \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\}$$

$$\Rightarrow \text{Span}\{\vec{y}_1, \dots, \vec{y}_k\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\}.$$

2). For any $j=1, 2, \dots, k-1$

$$\begin{aligned} (\vec{y}_k, \vec{y}_j) &= (\vec{x}_k - \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i, \vec{y}_j) \\ &= (\vec{x}_k, \vec{y}_j) - \left(\sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i, \vec{y}_j \right) \\ &= (\vec{x}_k, \vec{y}_j) - \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} (\vec{y}_i, \vec{y}_j) \\ &= (\vec{x}_k, \vec{y}_j) - \frac{(\vec{x}_k, \vec{y}_j)}{(\vec{y}_j, \vec{y}_j)} (\vec{y}_j, \vec{y}_j) \\ &= 0 \end{aligned}$$

$\Rightarrow \vec{y}_k$ is orthogonal to $\vec{y}_j$ for all $j=1, 2, \dots, j-1$.



Remarks: 1) The above construction from a linearly independent set $\{\vec{x}_1, \dots, \vec{x}_m\}$

to obtain an orthogonal set $\{\vec{y}_1, \dots, \vec{y}_m\}$ is called Gram-Schmidt.

2). $\{\vec{x}_1, \dots, \vec{x}_m\}$ $\xrightarrow{\text{G-S}}$ $\{\vec{y}_1, \dots, \vec{y}_m\}$ $\xrightarrow{\text{normalization}}$ $\{\vec{u}_1, \dots, \vec{u}_m\}$
 linearly independent orthogonal orthonormal.

3). Every finite-dimensional vector space has an orthonormal basis.

4). If $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is linearly dependent, then one of the $\vec{y}_j$'s

will be zero. In particular, suppose l is the smallest number such that $\{\vec{x}_1, \dots, \vec{x}_l\}$ is linearly independent, but $\{\vec{x}_1, \dots, \vec{x}_{l+1}\}$

is linearly dependent. Suppose we have constructed $\{\vec{y}_1, \dots, \vec{y}_l\}$

from G-S. Then $\vec{y}_{l+1} = \vec{x}_{l+1} - \sum_{i=1}^l \frac{(\vec{x}_{l+1}, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i$

As $\vec{x}_{l+1} \in \text{Span}\{\vec{x}_1, \dots, \vec{x}_l\} = \text{Span}\{\vec{y}_1, \dots, \vec{y}_l\}$, by the previous

lemma, $\vec{x}_{l+1} = \sum_{i=1}^l \frac{(\vec{x}_{l+1}, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i$. Thus $\vec{y}_{l+1} = \vec{0}$.

QR-Factorization: Let $A = (\vec{x}_1 \vec{x}_2 \dots \vec{x}_m) \in F^{n \times m}$ such that $\{\vec{x}_1 \vec{x}_2 \dots \vec{x}_m\}$ is linearly independent. Then there exists $Q = (\vec{u}_1 \vec{u}_2 \dots \vec{u}_m) \in F^{n \times m}$ with orthonormal columns and $R \in F^{m \times m}$ an upper triangular matrix with positive diagonal entries such that $A = QR$.

Proof: By G-S, take $\vec{y}_1 = \vec{x}_1 \Leftrightarrow \vec{x}_1 = \vec{y}_1$
 $\vec{y}_2 = \vec{x}_2 - \frac{(\vec{x}_2, \vec{y}_1)}{(\vec{y}_1, \vec{y}_1)} \vec{y}_1 \Leftrightarrow \vec{x}_2 = \frac{(\vec{x}_2, \vec{y}_1)}{(\vec{y}_1, \vec{y}_1)} \vec{y}_1 + \vec{y}_2$
 $\vdots$
 $\vec{y}_k = \vec{x}_k - \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i \Leftrightarrow \vec{x}_k = \sum_{i=1}^{k-1} \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)} \vec{y}_i + \vec{y}_k$
 $\vdots$

For each $k=2, \dots, m$: $a_{ik} = \frac{(\vec{x}_k, \vec{y}_i)}{(\vec{y}_i, \vec{y}_i)}$ for all $i=1, \dots, k-1$.

Then: $\vec{x}_k = (\vec{y}_1 \vec{y}_2 \dots \vec{y}_{k-1} \vec{y}_k \dots \vec{y}_m) \begin{pmatrix} a_{1k} \\ a_{2k} \\ \vdots \\ a_{k-1,k} \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

$$\Rightarrow (\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_m) = (\vec{y}_1 \ \vec{y}_2 \ \dots \ \vec{y}_m) \begin{pmatrix} 1 & a_{12} & a_{13} & \dots & a_{1m} \\ 0 & 1 & a_{23} & \dots & a_{2m} \\ 0 & 0 & 1 & \dots & a_{3m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Take $\vec{u}_k = \frac{\vec{y}_k}{\|\vec{y}_k\|}$, then $\{\vec{u}_1, \dots, \vec{u}_m\}$ is orthonormal

$$\text{and: } (\vec{y}_1 \ \vec{y}_2 \ \dots \ \vec{y}_m) = (\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_m) \begin{pmatrix} \|\vec{y}_1\| & & \\ & \|\vec{y}_2\| & \\ & & \ddots \\ & & & \|\vec{y}_m\| \end{pmatrix}$$

$$\begin{aligned} \Rightarrow (\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_m) &= (\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_m) \begin{pmatrix} \|\vec{y}_1\| & & \\ & \|\vec{y}_2\| & \\ & & \ddots \\ & & & \|\vec{y}_m\| \end{pmatrix} \begin{pmatrix} 1 & a_{12} & \dots & a_{1m} \\ 0 & 1 & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \\ &= \underbrace{(\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_m)}_{Q} \underbrace{\begin{pmatrix} \|\vec{y}_1\| & \|\vec{y}_1\| a_{12} & \dots & \|\vec{y}_1\| a_{1m} \\ 0 & \|\vec{y}_2\| & \dots & \|\vec{y}_2\| a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \|\vec{y}_m\| \end{pmatrix}}_{R} \end{aligned}$$

R , upper triangular

diagonals: $\|\vec{y}_1\| > 0, \dots, \|\vec{y}_m\| > 0$.



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